

L^AT_EX Crash Course

GRACE Transferable Skills

Dr. Anthea Alberto Dr. Ina Serif

12.05.2026

09:00-13:00

Intro

Welcome!

Agenda - 12.05.2026

- 09:00-09:30 Introductory round
- 09:30-09:45 Pros & cons of using $\text{\LaTeX}$
- 09:45-10:45 Structuring documents, syntax etc.
- 10:45-11:15 Break
- 11:15-11:45 Typography
- 11:45-12:15 Exercise
- 12:15-12:30 Discussing the exercise
- 12:30-13:00 Typesetting

Some smaller exercises will also be sprinkled into the program before the bigger one around 11:45.

Materials

You can find all the materials on our [GitHub repository](#).

Not everything is there yet, but you'll have full access to the slides, exercises and solutions by the end of the course.

About Me



- ▶ MA in Political Science from University of Zurich
- ▶ PhD on political communication and coalition governments from Humboldt-Universität zu Berlin
- ▶ Focus on quantitative text analysis
- ▶ Currently research navigator/research support for [RISE](#) at Uni Basel

Introductory Round

Please tell us a bit about yourself:

- ▶ Name
- ▶ Field of study
- ▶ Special requirements from your field (equations, specific typography etc.)

What is L^AT_EX?

- ▶ In simple terms: it's a typesetting software
- ▶ Main difference to Word et al.: what you see is not what you get
- ▶ LaTeX uses a plain text editor that later gets compiled into a nicely typeset PDF
- ▶ LaTeX operates with *commands* and *environments* - things you will learn about today

Where and how can I use L^AT_EX?

This course uses [Overleaf](#), an online LaTeX editor, as its main teaching tool.

The benefit of Overleaf is that you don't have to install anything, you just need to make an account (a plus if you use a managed laptop).

Plus, it offers a lot of features and is well suited for working on solo projects.

Where and how can I use L^AT_EX?

The drawbacks of Overleaf are the online requirement and the fact that some features require a subscription.

Unfortunately there's no campus license, but inquire about potential group licenses with your supervisor, or the possibility of reimbursement.

Where and how can I use LaTeX?

Many editors exist that allow working with LaTeX while being offline.

I have used [TeXstudio](#) and [MiKTeX](#) as package manager, but not recently.



TeXlyre

TeXlyre is a free, open-source & local-first LaTeX Collaborative Web Editor.

First attempts at using it have been promising, but it comes with some caveats:

- ▶ If you empty your browser cache without having done a backup, your projects will be gone
- ▶ Multiple ways of backing up your projects exist, but not all are equally easy to set up
- ▶ Unclear how many collaborators are supported

Important: there rarely is “one true way” of doing things in LaTeX. This course aims to get you started and to teach you good practices.

LaTeX and AI/LLMs

Due to the highly formalized nature of LaTeX, LLMs tend to do quite well with it.

I can recommend using LLMs when you don't know how to fix a problem, but I generally advise against getting them to create whole documents from scratch, particularly if you want to actually learn something.

I still prefer to use a search engine when I'm stuck, and to look for potential solutions on StackOverflow et al. It forces you to really think about the problem.

Why you should (not) learn to use $\text{\LaTeX}$:
a non-comprehensive list

Pro

L^AT_EX...

- ▶ ...focuses on **content**, not layout.
- ▶ ...makes **bibliographies** easier to handle.
- ▶ ...has many **libraries and packages** that make life easier.
- ▶ ...is customizable and flexible.
- ▶ ...has a **big user base** and many tutorials.
- ▶ ...makes it easier to include **mathematical notation**.

Big user base

The screenshot shows the TeX Stack Exchange homepage. At the top, there's a navigation bar with the TeX logo and a search bar. Below the navigation bar, there's a sidebar on the left with links to Home, Questions, Unanswered, Tags, Chat, Users, and Companies. The main content area is titled "Explore our questions" and lists several questions with their respective details, including votes, answers, and views. The questions are:

- Aligning LaTeX inequality at Multiple Points (3 votes, 3 answers, 171 views)
- How to invoke the beta variant in the XITS-Italic font? (1 vote, 0 answers, 3 views)
- I want to have a custom table of contents for each section but subentries don't appear as desired (0 votes, 1 answer, 23 views)
- Typeset a matrix with crossing lines (5 votes, 4 answers, 161 views)
- Script - how to add chapter and section in header when sections appear, otherwise chapter only (2 votes, 0 answers, 10 views)

On the right side, there's a section titled "Hot Network Questions" which lists several trending questions, including:

- Is it possible to directly check for source code modification?
- Is it possible to avoid nesting TikZ environments in an RST environment?
- Adult files - "disappear"
- Retrospective copyright violation
- Soldering a FLO-22 socket on a PCB
- Why is there no pull up or pull down resistor for a 5VAC driver circuit?
- Typeset a matrix with crossing lines
- Comedy about super-additive bread of beer
- Theoretical possibilities of flight capable life, before land animals
- How to obtain cytoplasm from L4-hermaphrodite?
- What do the two angles on the Ark of the Covenant represent?
- Resistance Axiom - any strange better than staying silent?
- What would I need to build an AI agent to write good latex documents?
- Should "Center Level of the Commoner" actually be "Bumsteiner" in West German?
- How to specify fontsize in PySCT?
- How are languages managed during the compile?
- Checking for One-Time Passwords
- What's average latency, up/down/uplinks for...

Screenshot taken on April 24th 2025, 16:15

Contra

I would not be teaching this course if I thought you *shouldn't* actually learn LaTeX.

Therefore, the next section is not necessarily about arguments *against* learning it, but rather some *caveats*; or things to keep in mind.

Contra

- ▶ There is a **steep learning curve** and **potential for frustration**, particularly at the beginning.
- ▶ What you see is **not** what you get (unlike Word et al.)
- ▶ “More options” sometimes comes at the expense of efficiency (plus the risk for “dependency graveyards”).
- ▶ **Version control** is not always straightforward.
- ▶ Obstacles to working on a document collaboratively/simultaneously.
- ▶ Potential co-authors might not be familiar with LaTeX.

WYSINWYG

```

58 a non-comprehensive list
59 \end{frame}
60
61 \begin{frame}{Pro}
62 \LaTeX...
63 \vspace{2mm}
64 \begin{itemize}
65   \item ...focuses on content, not layout.
66   \pause
67   \item ...makes bibliographies easier to handle.
68   \pause
69   \item ...has many libraries and packages that make life easier.
70   \pause
71   \item ...is customizable and flexible.
72   \pause
73   \item ...has a big user base and there are many tutorials.
74   \pause
75   \item ...makes it easier to include mathematical notation.
76 \end{itemize}
77 \end{frame}
78
79 \begin{frame}{Big user base}

```

Pro

LaTeX...

- ▶ ...focuses on content, not layout.
- ▶ ...makes bibliographies easier to handle.
- ▶ ...has many libraries and packages that make life easier.
- ▶ ...is customizable and flexible.
- ▶ ...has a big user base and there are many tutorials.
- ▶ ...makes it easier to include mathematical notation.

What you see is not (directly) what you get.

WYSINWYG

However, Overleaf has a Rich Text option that approaches WYSIWYG and offers more Office-style editing options.



Contra

- There is a **steep learning curve** and **potential for frustration**, particularly at the beginning.
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- **Version control** is not always straightforward.
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- Potential co-authors might not be familiar with LaTeX.

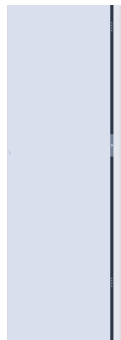
WYSINWYG



WYSINWYG

However, Overleaf has a Rich Text option that approaches WYSIWYG and offers more Office-style editing options.

```
\vspace{1.5mm}
```



Version Control

- ▶ For Overleaf free plan: [only last 24 hours](#) available.
- ▶ Syncing with git, GitHub or Dropbox is possible for premium users.
- ▶ For other editors (e.g. TeXstudio), versioning is possible via Dropbox, [GitHub](#) etc.
- ▶ TeXstudio has its own version control system/git integration, but it needs to be set up first.

Working collaboratively

- ▶ Limited options on Overleaf's free plan: **only one collaborator per project**
- ▶ For other editors: similar problems as with version control; no straightforward option to work on a document simultaneously
- ▶ Possibility to use external software to facilitate collaboration
- ▶ Again, maybe worth it to keep an eye on TeXlyre...

Structuring L^AT_EX documents

Some terminology & notation:

- ▶ *Source file* or *.tex file*: what you're working on, not the compiled file
- ▶ *Commands* start with a backslash (on Swiss keyboards: press option/alt + Shift + 7)
- ▶ If a command takes an argument, it's in *curly brackets*
- ▶ You'll see a lot of examples of this today

Structure of a L^AT_EX document

preamble and body

preamble defines global options for layout, font, bibliography asf.

body contains the actual text

minimal example:

<code>\documentclass[10pt,a4paper]{article}</code>	← <i>preamble</i>
<code>\begin{document}</code>	
Some meaningful text.	← <i>body</i>
<code>\end{document}</code>	

Structure of a $\text{\LaTeX}$ document

Some meaningful text.

Structure of a $\text{\LaTeX}$ document

preamble, body and top matter/title page

minimal example:

```

\documentclass[10pt,a4paper]{article}      ← preamble
\begin{document}
\title{Title of the document}             ← topmatter
\author{Arthur Dent}
\date{\today}
\maketitle
Some meaningful text.                      ← body
\end{document}

```

Structure of a $\text{\LaTeX}$ document

Title of the document

Arthur Dent

April 25, 2023

Some meaningful text.

Structuring L^AT_EX documents

chapters, sections, paragraphs as common document structures

```
\chapter{Chapter name}  
\section{Header name on level 1}  
\subsection{Header name on level 2}  
\subsubsection{Header name on level 3}  
\paragraph{Paragraph name}  
\subparagraph{Subparagraph name}
```

Structuring L^AT_EX documents

Note: All structural elements will be numbered automatically:

1 A section of the document

Some meaningful text.

1.1 A subsection

1.1.1 Followed by a subsection

More meaningful text.

1.2 Another subsection

A very interesting fact.

2 Back to section level

And even more meaningful text.

You can leave subsections unnumbered by using `\setcounter`:

```
\setcounter{secnumdepth}{1}
```

Structuring L^AT_EX documents

You can leave certain chapters/sections unnumbered, using “*”.
This will also not include the section into the table of contents:

```
\section*{Acknowledgements}
```

1 A section of the document

Some meaningful text.

1.1 A subsection

1.1.1 Followed by a subsubsection

More meaningful text.

1.2 Another subsection

A very interesting fact.

2 Back to section level

And even more meaningful text.

Acknowledgements

This does not need to be numbered.

Try it yourself!

Structuring L^AT_EX documents

As with all default settings in L^AT_EX:

You can change the behaviour of everything, e.g. vertical space between headers and text, but think twice if it's necessary.

You can insert comments that explain modifications, or just to make a note to yourself, or to keep a paragraph in a document, but without typesetting it, using "%".

This is a very good thought. % look up missing
reference!

Syntax, commands, environments and packages

How to

- ▶ write commands
- ▶ change default behaviour
- ▶ use environments
- ▶ use packages

Basic syntax and commands

Basic syntax:

`\ + command`

`\ + command + {mandatory argument}`

`\ + command + [optional argument] , + {mandatory argument}`

Basic syntax and commands

Basic syntax:

inserting page break:

`\newpage`

changing font:

`\textit{italic text}` → *italic text*

inserting a section:

`\section{Section title}` → **1 Section title**

defining document class:

`\documentclass[10pt,a4paper]{article}` →
a document with default settings according to class "article", with
changes regarding font size and paper format

Try it yourself - go back to your example document and play around with the syntax and settings for a bit.

Environments

environments are used to apply commands to a defined section of a document

use existing or define new environments

environments start with `begin{name}` and end with `end{name}`

```
\begin{itemize}
  \item Item 1   → • Item 1
  \item Item 2   • Item 2
\end{itemize}
```

Try it yourself!

Packages

packages contain additional LaTeX commands to change style features or to modify existing ones

most LaTeX distributions, such as MiKTeX, include a package manager

packages can also be installed via the command line

packages are called in the preamble:

```
\usepackage[english]{babel}
```

Packages

packages are called in the preamble:

```
\usepackage[paper=letterpaper,  
marginparwidth=3in, % Length of section titles  
marginparsep=-3in, % Space between titles and text  
margin=1in, % 1 inch margins  
includemp] % includes the margin notes  
{geometry}
```


Typography primer

- ▶ font style and size
- ▶ quotes, citations, footnotes
- ▶ in-text referencing

Font style and size

font style:

`\underline{This}` is a `\emph{text}` with a `\textbf{lot}`
of different `\textsc{styles}`.

This is a *text* with a **lot** of different STYLES.

Font style and size

font size:

`{\tiny This}` is a `{\Huge text}` with a `{\Large lot}` of
different `{\LARGE sizes}`.

This is a **text** with a lot of different sizes.

quotes, citations and footnotes

```
\begin{quote}  
  quoted text.\footnote{footnote text}  
\end{quote}
```

This is a longer quote from a scientific article that I would like to cite in its whole beauty, including the reference.¹

¹Adams, Douglas: The Hitchhiker's Guide to the Galaxy. London 1979, p. 42.

Try it yourself!

quotes, citations and footnotes

How to use your reference manager (Zotero, Citavi, ...) with LaTeX to create a bibliography will be covered next session – stay tuned!

in-text referencing

you can mark structural, textual or graphic elements in a document and reference to it

you can label a section or a figure:

```
\section{Great section title}\label{great}
```

(figures next session!)

As discussed in `\ref{great}`, I will now ...

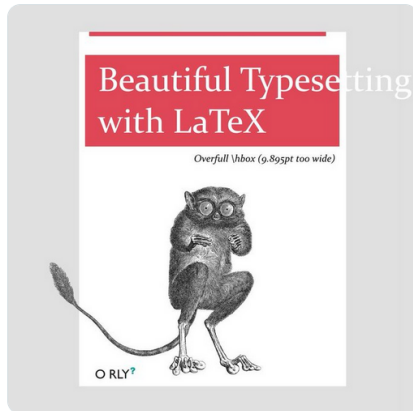
As discussed in chapter 2, I will now ...

Exercise



Astro Joke of the Week @JokesAstro · 23. Apr.

I didn't do a LaTeX joke in a while.



14



133



1.288



40.235



[Diesen Thread anzeigen](#)

Underfull/overfull boxes

Generally not a huge issue, unless you prepare something for publication (like your dissertation). In that case, be mindful of overfull boxes.

Some workarounds: see [here](#) and [here](#) on Stack Exchange.

There are some packages and commands that help get rid of these problems.

You can also remove automatic hyphenation and hyphenate by hand, or reformulate the sentence.

documentclass

At the beginning of each LaTeX file, you have to specify what *type* of document you want to create.

This happens with the *documentclass* command.

documentclass also lets you set additional options for your document.

These slides uses a very simple form of document class:

`\documentclass{beamer}`, where what's in the curly brackets indicates the document type.

Frequently used arguments for `\documentclass{}`:

- ▶ article
- ▶ beamer
- ▶ report
- ▶ book or scrbook
- ▶ letter

More options

In addition to type, you can add font size, paper size and format and more to costumize your document.

This is done via square brackets before the curly ones.

For example:

```
\documentclass[12pt,a4paper,oneside]{scrbook}
```

Standard options (taken from [texblog](#)) include:

- ▶ Font size (10pt, 11pt, 12pt...)
- ▶ Paper size and format (a4paper, letterpaper...)
- ▶ Multiple columns (onecolumn, twocolumn)
- ▶ Title page behavior (titlepage, notitlepage)
- ▶ Draft mode (draft)

Most of these options and costumizations can also be done with packages like *geometry*, as you've seen in an earlier slide.

Example:

```
\usepackage[a4paper,top=2cm,bottom=2cm,left=3cm,  
right=3cm,marginparwidth=1.75cm]{geometry}
```

Themes

For presentations like this one, you can use a *theme* for your slides.

This presentation uses the *Szeged* theme; you call it by putting `\mode<presentation> {\usetheme{Szeged}}` into the preamble.

You can find a gallery of themes [here](#). There are many more options, and you can of course make your own.

Some animation effects à la Powerpoint are possible, but not very straightforward to implement.

Language options

Set the language: either within `documentclass`, or with the *babel* package.

These slides use `\usepackage[english]{babel}`.

You can use multiple languages within the same document, just separate them with a comma.

Language options

Babel supports various languages (documentation [here](#) and [here](#)), but if you require a non-Latin alphabet (like Arabic, Hebrew, Japanese, Mandarin...) the situation is unfortunately a bit more complicated.

You can find an intro on using the [polyglossia](#) package with Overleaf [here](#).

End of day 1

Are there any questions?